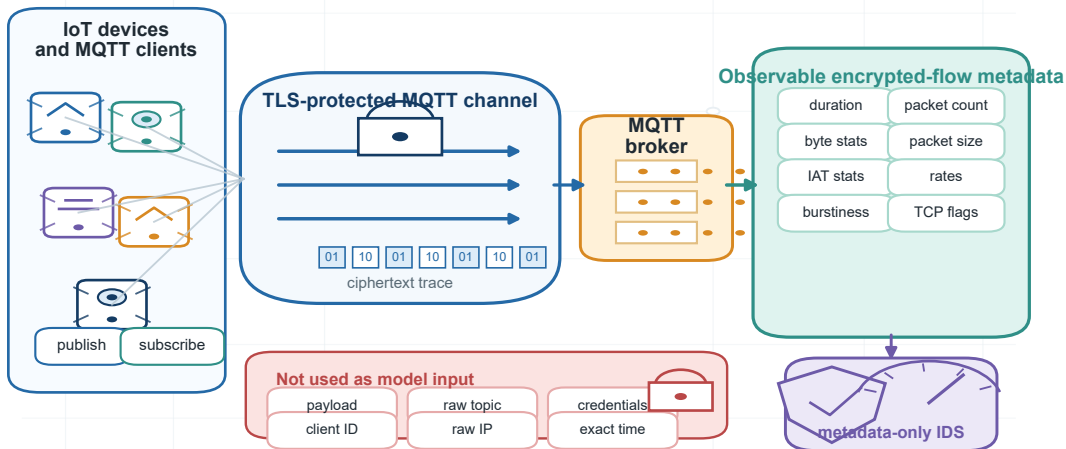


Encrypted MQTT observability boundary

What a deployable metadata-only IDS can observe without decrypting TLS or collecting sensitive raw fields



Bounded claim: detection layer only, not cryptographic authentication or protocol hardening.